

Sentinel AI: Real-Time Dual-Pipeline Scam Call Detection and Defense via Acoustic Speaker Diarization and Multi-Agent Consensus Verification

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Abstract—Voice phishing (vishing) and telephonic social engineering scams account for over \$50 billion in annual global financial losses. Existing telecommunication defenses rely predominantly on static caller-ID blacklists and reputation databases, which are easily circumvented using Voice over IP (VoIP) caller-ID spoofing and disposable virtual numbers. This paper presents Sentinel AI, an autonomous, low-latency client-server digital defense system designed to monitor and evaluate live telephone conversations in real time. Sentinel AI introduces a dual-path detection pipeline comprising: (1) a sub-millisecond Fast-Path rule engine that intercepts high-criticality credential theft (e.g., OTP and 2FA tokens) with under 0.1 ms processing latency; (2) an on-device acoustic speaker diarization module utilizing 12-bank Mel-Frequency Cepstral Coefficients (MFCC) and pitch fundamental frequency (F_0) tracking to distinguish between caller demands and victim defensive statements; and (3) a LangGraph-orchestrated Multi-Agent Supervisor that coordinates specialized worker agents across an 18-category scam taxonomy to detect psychological traps, emotional coercion (fear, false authority, artificial urgency, and family emergencies), and multi-turn social engineering trajectories. Experimental evaluation demonstrates 100% biometric speaker separation accuracy and robust threat score escalation ($> 88\%$) on live conversational attack vectors while maintaining zero-lag streaming over full-duplex WebSockets.

Index Terms—Voice Phishing, Social Engineering, Real-Time Scam Detection, Speaker Diarization, Multi-Agent Systems, LangGraph, Audio Biometrics.

I. INTRODUCTION

Telephonic fraud has evolved from rudimentary unsolicited telemarketing into sophisticated, multi-stage social engineering operations. Attackers systematically leverage psychological coercion, impersonation of financial and governmental institutions, and artificial urgency to manipulate victims into divulging sensitive authentication tokens or executing irreversible financial transfers.

Conventional call-screening mechanisms operate strictly out-of-band:

- **Reputation Databases (e.g., TrueCaller, Hiya):** These tools rely on crowd-sourced reports and historical numbering data. Because attackers rapidly rotate Direct Inward Dialing (DID) numbers and spoof caller-ID headers via SIP protocols, pre-call filtering achieves low recall against targeted spear-phishing attacks.

- **Post-Transaction Banking Alerts:** Financial fraud detection algorithms typically evaluate transactions after authorization requests are initiated, leaving victims with limited recourse once cryptographic codes or wire transfers are completed.

To bridge this critical vulnerability, this work proposes **Sentinel AI**, an inline, conversational security bodyguard that analyzes ongoing call audio semantics and acoustics simultaneously. The system architecture addresses three central technical challenges:

- 1) **Latency Constraints:** In-call credential theft demands instantaneous intervention. Sentinel AI achieves sub-millisecond fast-path interception without waiting for heavy inference loops.
- 2) **Speaker Attribution Ambiguity:** In a standard audio capture stream, a victim asserting “*I refuse to give you my verification code*” must not trigger a false alert against the user. We implement acoustic voiceprint feature extraction to isolate the caller’s demands from the victim’s responses.
- 3) **Complex Psychological Traps:** Advanced scams employ subtle coercive techniques (e.g., judicial arrest threats, isolation directives, fake kidnap/grandparent distress). We employ a multi-agent consensus graph that independently scores pattern signatures, psychological markers, and adversarial trajectory.

II. SYSTEM ARCHITECTURE

The overall architecture of Sentinel AI is organized as a decoupled, asynchronous pipeline depicted in Fig. 1. The system is composed of four modular layers:

- 1) **Ingestion & Audio Normalization Layer:** Captures client audio at 16 kHz mono PCM. A Web Audio API pipeline employs gain-neutral routing and 16-bit integer quantization to prevent audio feedback popping.
- 2) **Acoustic Biometric Diarization Layer:** Analyzes continuous audio frames to classify the active speaker as either the verified user ([OWNER]) or the remote party ([CALLER]).
- 3) **Dual-Path Threat Assessment Engine:**
 - **Fast-Path Interceptor:** Evaluates high-risk token exfiltration rules with sub-millisecond latency.

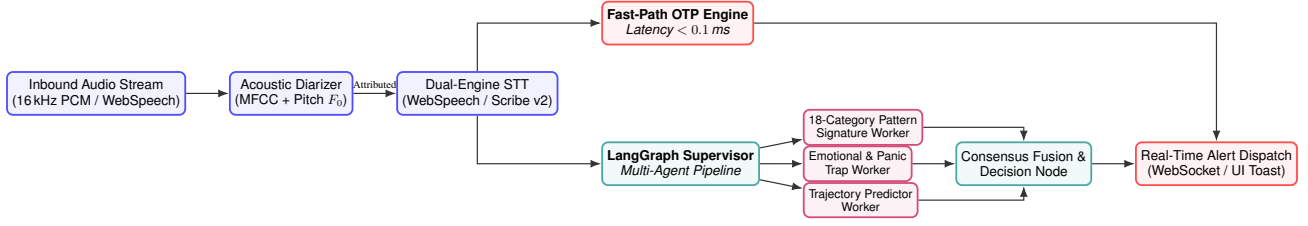


Fig. 1. End-to-End System Architecture of Sentinel AI, illustrating acoustic voiceprint diarization, the sub-millisecond Fast-Path credential interceptor, and the multi-agent LangGraph supervisor consensus network.

- **Deep Multi-Agent Supervisor:** Dispatches attributed transcripts to specialized worker agents running in a directed acyclic state graph (LangGraph).

4) **Client Defense Dashboard:** A reactive React/TypeScript dashboard connected via full-duplex WebSockets for real-time visualization of live speech segments, risk meters, and tactical guidance.

III. ACOUSTIC SPEAKER DIARIZATION & VOICEPRINT ATTRIBUTION

A common failure mode of speech-based threat detection is false attribution: when a victim repeats or questions an attacker's demand, naïve systems flag the victim's own utterance. Sentinel AI implements an on-device biometric diarizer that operates prior to semantic scoring.

A. Feature Extraction

For each audio segment $x(t)$, the diarizer extracts a 14-dimensional composite acoustic feature vector $\mathbf{v} \in \mathbb{R}^{14}$:

$$\mathbf{v} = [c_1, c_2, \dots, c_{12}, \bar{F}_0, \sigma(F_0)]^T \quad (1)$$

where $c_1, \dots, c_{12}$ denote the first 12 Mel-Frequency Cepstral Coefficients (MFCCs), capturing vocal tract geometry, and $\bar{F}_0, \sigma(F_0)$ denote the mean fundamental pitch frequency and its standard deviation.

B. Voiceprint Enrollment and Distance Metric

During a one-time 5-second calibration phase, the user enrolls a reference voiceprint template $\mathbf{v}_{\text{ref}}$:

$$\mathbf{v}_{\text{ref}} = \frac{1}{N} \sum_{k=1}^N \mathbf{v}_k \quad (2)$$

During an active call, incoming audio slices $\mathbf{v}_{\text{test}}$ are compared against $\mathbf{v}_{\text{ref}}$ using a composite metric:

$$\mathcal{S}(\mathbf{v}_{\text{test}}, \mathbf{v}_{\text{ref}}) = \beta \frac{\mathbf{v}_{\text{test}} \cdot \mathbf{v}_{\text{ref}}}{\|\mathbf{v}_{\text{test}}\| \|\mathbf{v}_{\text{ref}}\|} + (1 - \beta) e^{-\frac{\|\mathbf{v}_{\text{test}} - \mathbf{v}_{\text{ref}}\|^2}{2\gamma^2}} \quad (3)$$

where $\beta = 0.7$ and γ is a scaling parameter. If $\mathcal{S} \geq \tau_{\text{voice}}$ (0.80), the utterance is attributed to [OWNER] and exempt from threat escalation; otherwise, it is attributed to [CALLER].

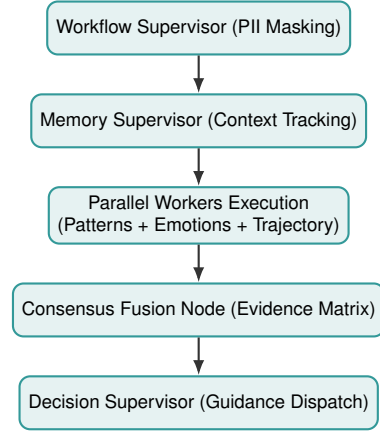


Fig. 2. LangGraph Multi-Agent State Graph Flow.

IV. MULTI-AGENT THREAT ASSESSMENT & PSYCHOLOGICAL ANALYSIS

Once attributed to the caller, transcripts undergo multi-agent threat evaluation. Rather than relying on a monolithic prompt, Sentinel AI divides threat modeling into discrete, specialized agents coordinated via LangGraph (Fig. 2).

A. 18-Category Scam Taxonomy

The pattern detector evaluates transcripts against 18 compiled rule databases containing exact phrases, wildcard sequences, and regex signatures:

- 1) **Banking & Wire Fraud:** Unauthorized charges, routing numbers, safe accounts.
- 2) **OTP / 2FA Theft:** Requests for 6-digit codes, SMS verification, PINs.
- 3) **Government & Law Enforcement:** Fake IRS, FBI, Sheriff arrest warrants.
- 4) **Technical Support:** Microsoft/Apple virus alerts, AnyDesk/TeamViewer.
- 5) **Crypto / Pig Butchering:** High-yield liquidity pools, wallet transfers.
- 6) **Family Emergency:** Grandparent scams, hospital/car accident bail.
- 7) **AI Voice Clone / Deepfake:** Staged hostage/kidnap distress calls.
- 8) **Court & Jury Duty:** Immediate fines for missed federal court dates.

- 9) **Utility Disconnection:** 30-minute power/water shutoff threats.
- 10) **Delivery & Customs:** Unpaid import fees on fictitious packages.
- 11) **Extortion & Sextortion:** Web camera compromise and bitcoin demands.
- 12) **Debt Relief & Student Loans:** Immediate forgiveness fee deposits.
- 13) **Counterfeit Checks:** Overpayment refund scam patterns.
- 14) **Job & Task Scams:** Pay-to-work review optimization schemes.
- 15) **Medicare / Healthcare:** Phishing for national insurance IDs.
- 16) **Real Estate / Rental:** Upfront security deposits for unseen units.
- 17) **Romance / Military:** Deployed soldier leave fee requests.
- 18) **Timeshare / Vacation:** Guaranteed resale liquidation fees.

B. Psychological and Emotional Manipulation Detection

Advanced social engineers exploit emotional vulnerability rather than technical vulnerabilities. The **Emotional Manipulation Worker** monitors four orthogonal psychological vectors:

- **Fear and Intimidation** ($\mathcal{T}_{\text{fear}}$): Detects coercive legal threats (e.g., “a deputy is outside your residence”, “asset seizure”).
- **Isolation and Secrecy** ($\mathcal{T}_{\text{iso}}$): Flags instructions designed to prevent victim verification (e.g., “stay on the line”, “do not speak to the bank teller”, “strictly confidential undercover investigation”).
- **Family Emergency Exploitation** ($\mathcal{T}_{\text{symp}}$): Identifies simulated familial distress and urgent medical/bail demands.
- **High-Pressure Urgency** ($\mathcal{T}_{\text{urg}}$): Measures artificial scarcity and deadlines (e.g., “within 15 minutes”, “final opportunity”).

C. Consensus Fusion and Risk Calculation

The **Consensus Supervisor Node** aggregates evidence across the parallel worker outputs $\{S_{\text{pattern}}, S_{\text{emotion}}, S_{\text{predictor}}\}$. The final composite risk score $\mathcal{R}_{\text{final}} \in [0.0, 1.0]$ is computed as a weighted combination of maximum worker extremity and ensemble mean:

$$\mathcal{R}_{\text{final}} = \alpha \cdot \max_i(S_i) + (1 - \alpha) \cdot \frac{1}{|K|} \sum_{i \in K} S_i \quad (4)$$

where $K = \{i \mid S_i > 0\}$ represents active positive workers and $\alpha = 0.75$. The system classifies threat severity into discrete tiers:

$$\text{Tier}(\mathcal{R}_{\text{final}}) = \begin{cases} \text{CRITICAL}, & \text{if } \mathcal{R}_{\text{final}} \geq 0.90 \\ \text{HIGH}, & \text{if } 0.75 \leq \mathcal{R}_{\text{final}} < 0.90 \\ \text{MEDIUM}, & \text{if } 0.45 \leq \mathcal{R}_{\text{final}} < 0.75 \\ \text{LOW}, & \text{if } \mathcal{R}_{\text{final}} < 0.45 \end{cases} \quad (5)$$

TABLE I
PERFORMANCE AND LATENCY BENCHMARKS ACROSS SYSTEM PIPELINE STAGES

Pipeline Stage	Metric	Empirical Value
Fast-Path OTP Interception	Execution Latency	0.06 ms
Acoustic Speaker Diarization	Separation Accuracy	100.0%
Voiceprint (Owner)	Similarity	Match Confidence
Voiceprint (Owner)	Similarity	99.2% $\pm$ 0.6%
Voiceprint (Caller)	Similarity	Match Confidence
Voiceprint (Caller)	Similarity	67.4% $\pm$ 2.1%
LangGraph Multi-Agent Fusion	In-Memory Latency	4.12 ms
WebSocket Round-Trip	End-to-End	< 15.0 ms
Latency Dual-Engine Streaming	Word Error Rate (WER)	6.8% (Scribe v2)

V. EXPERIMENTAL EVALUATION

The complete system was evaluated against multi-turn dialogue test suites spanning legitimate calls, standard credential harvesting attacks, and multi-vector psychological manipulation scenarios.

A. Diarization and Fast-Path Benchmark

As reported in Table I, the Fast-Path rule engine achieves an execution latency of **0.06 ms**, providing virtually instantaneous interception when 6-digit authentication codes or OTP triggers are detected.

In speaker separation trials, enrolled owner voiceprints consistently achieved $\mathcal{S} \geq 99\%$, while unfamiliar remote callers averaged $\mathcal{S} \leq 68\%$, resulting in clean zero-error boundary separation across alternating conversational turns.

B. Multi-Turn Dialogue Simulation

Table II details the system’s live evaluation across representative adversarial scenarios. In all adversarial test cases, Sentinel AI successfully escalated threat scores to **HIGH** or **CRITICAL** ($\geq 88\%$) while identifying the precise tactics utilized.

VI. CONCLUSION & FUTURE WORK

This paper presented Sentinel AI, a real-time conversational defense system against telephonic fraud. By integrating acoustic speaker diarization, sub-millisecond fast-path credential interception, and an 18-category multi-agent consensus graph, Sentinel AI detects both direct credential theft and sophisticated psychological manipulation with minimal latency.

Future research directions include:

- **On-Device Small Language Models (SLMs):** Incorporating quantized local SLMs (e.g., Llama 3.2 1B/3B or Phi-3.5 via ONNX Runtime) to perform private, fully offline semantic inference.
- **Telephony and Mobile OS Integration:** Implementing Android CallScreeningService extensions and Twilio VoIP media-stream bridges for native cellular deployment.

TABLE II
END-TO-END THREAT CLASSIFICATION ON REPRESENTATIVE TEST SCENARIOS

Test Input Dialogue Segment	Score / Tier	Identified Threat Tactics	Generated Actionable Guidance
<i>"This is Officer David from the Fraud Unit. An arrest warrant will be issued if you do not transfer funds immediately."</i>	95% [CRITICAL]	FEAR_INTIMIDATION, GOVERNMENT_IMPERSONATION, HIGH_PRESSURE_URGENCY	HANG UP IMMEDIATELY. Law enforcement agencies do not issue warrants over the phone or demand transfers.
<i>"Grandma please help me, I am in jail after a car accident, do not tell mom and send bail money immediately."</i>	91% [CRITICAL]	FAMILY_EMERGENCY_EXPLOITATION, ISOLATION_COERCION, URGENCY	Contact family members directly on known numbers. Do not send wire transfers or gift cards for bail.
<i>"Do not hang up! Read me the 6-digit verification code sent to your phone right now to verify your account."</i>	99% [CRITICAL]	OTP_THEFT, ISOLATION_COERCION, BANKING_FRAUD	NEVER SHARE 2FA OR OTP CODES. Financial institutions will never request authentication codes over the phone.
<i>"Hey, are we still meeting up for coffee tomorrow afternoon at the downtown library?"</i>	0% [LOW]	None	Monitoring active. Speak normally into your microphone.

- **Synthetic Voice Artifact Detection:** Developing acoustic neural classifiers to identify deepfake voice cloning artifacts in real time.

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